

- (a)** Fig. 5.1 shows the primary structure of a section of a polypeptide chain of collagen.



..... [3]

- Complete the passage to describe the process of exocytosis.

[3]

- (c) Hydrolytic enzymes, known as collagenases, are secreted by cells in an inactive form.

Cells also secrete inhibitors of collagenases. The activity of the enzymes and inhibitors is regulated so that the development and maintenance of the extracellular matrix is controlled.

- (i) State **and** explain what the outcome will be for the composition of the extracellular matrix if collagenase inhibitor activity is needed.

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..... [2]

- (ii) Synthetic inhibitors have been trialled as potential treatment for diseases caused by a lack of regulation of collagenase activity.

Research involves investigating the mechanism of action of an inhibitor.

State the effect that a **non-competitive** inhibitor will have on the maximum rate of reaction,  $V_{\max}$ , and the Michaelis–Menten constant,  $K_m$ , of collagenase.

$V_{\max}$  .....

$K_m$  ..... [2]

[Total: 10]